

Name	Subject Class	Class	Candidate Number
	2BI		



ANGLO-CHINESE JUNIOR COLLEGE
Preliminary Examination 2018

BIOLOGY

HIGHER 2

9744/02
17 AUGUST 2018
2 hours

Paper 2 Structured Questions

READ THESE INSTRUCTIONS FIRST

Write your name, index number and class on the top of this page.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.
No additional materials are required.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
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9	
10	
Total	100

- 1 Cholesterol is synthesised in the smooth endoplasmic reticulum (SER) in liver cells by a series of enzyme-catalysed reactions. Cholesterol is then transported to the Golgi apparatus where they are packaged into vesicles and subsequently released into a membrane-bound duct of the liver.

Fig. 1.1 is an electron micrograph of a section of a liver tissue.



Fig. 1.1

- (a) Name structure **T** in Fig. 1.1 and describe its role in liver cells.

1. **Structure T is the mitochondrion;**

2. **Site of ATP synthesis for synthesis of cholesterol/glycogen (or other liver-specific functions);**

[2]

- (b) Both prokaryotes and structure **T** have membrane proteins to help them perform the role described in (a). Suggest how prokaryotes perform this role.

1. **Presence of electron carriers / ATP synthase;**

2. **Embedded in cell surface membrane;**

3. **Electron carriers use the energy from the transport of the electrons to pump H^+ across the membrane, generating a proton gradient/pool;**

4. **ATP synthase which uses the energy of the proton gradient/proton motive force/flow of protons for chemiosmotic synthesis of ATP;**

5. **AVP;**

[3]

- (c) Describe the role of cholesterol in the cell surface membrane.

Maintain fluidity of the membrane by preventing close packing of phospholipids at low temperature / prevent phospholipids from moving too far apart when temperature is high;

[1]

- (d) Suggest how cholesterol is transported from the Golgi apparatus to the membrane-bound duct of the liver.

1. **Secretory vesicles containing cholesterol pinch off the GA and move along microtubules to the cell surface membrane;**
2. **They fuse with the cell surface/duct membrane and discharge cholesterol outside the cell via exocytosis;**

[2]

[Total: 8]

- 2 Fig. 2.1 shows the activation of a G-protein linked receptor (GPLR) found in the cell surface membrane upon binding of a drug, salbutamol. Salbutamol is often used in the treatment of asthma – a long-term condition which narrows the airways in the lungs. The G protein consists of three subunits - α , β , and γ .

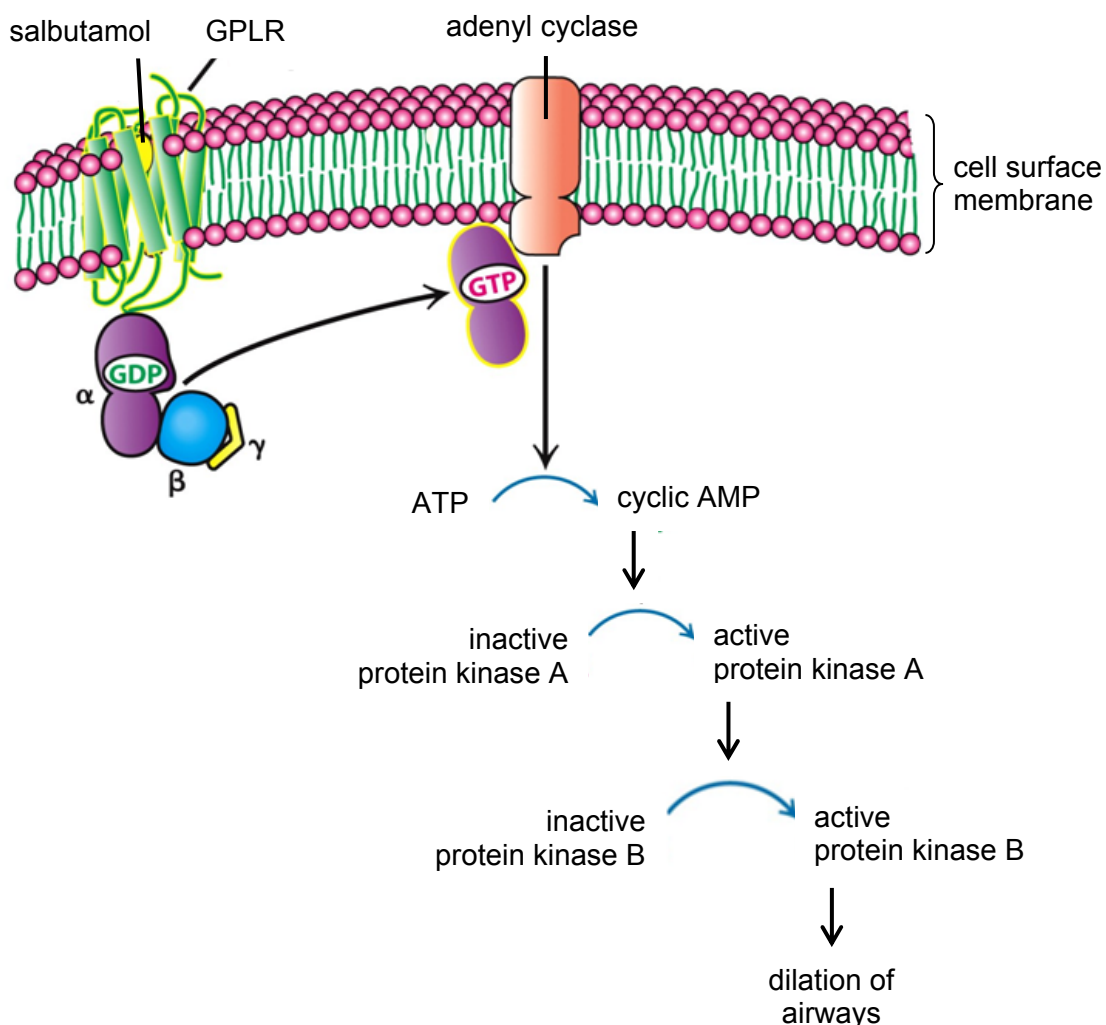


Fig. 2.1

(a) Describe how GPLR is held within the cell surface membrane.

- Hydrophobic interactions occur between the hydrophobic core/hydrocarbon tails of phospholipid bilayer and hydrophobic R groups of amino acid residues of GPLR;**
- Hydrogen/ionic bonds formed between charged/ hydrophilic phosphate heads of phospholipids and hydrophilic R groups of amino acids in cytoplasmic + extracellular regions of GPLR;**

[2]

(b) With reference to Fig. 2.1, describe how the use of salbutamol can relieve symptoms of asthma.

1. **Binding of salbutamol to a specific/complementary binding site of GPLR result in conformational change of receptor;**
2. **Activated GPLR activates G protein, where GTP replaces GDP;**
3. **α subunit dissociates from $\beta\gamma$ subunits;**
4. **α subunit binds to and activates adenylyl cyclase (at the membrane), which catalyses the conversion of ATP to cAMP;**
5. **cAMP act as second messengers and activates protein kinase A;**
6. **trigger a phosphorylation cascade resulting in amplification of signal, causing to dilation of airways;**

[4]

(c) There are many different types of receptors used by a cell to transduce extracellular signals. Some are located on the cell surface membrane while others are located within the cell.

Explain why some receptors are found on the cell surface membrane while others are found within the cell.

1. **Ligands which are lipid-soluble/hydrophobic/non-polar bind to specific nuclear receptors within the cell;**
2. **Ligands which are not lipid-soluble/hydrophilic/charged/polar bind to specific receptors on the membrane;**
3. **Ref to size;**

[2]

[Total: 8]

-

(a) (i) With reference to Fig. 3.1, describe how the structure of a tRNA molecule differs from the structure of a mRNA molecule.

@1m per valid comparison

(ii) Explain how these differences allow tRNA and mRNA molecules to perform their roles.

@1m per structure-function relationship

Fig. 3.2 shows a tRNA molecule that is able to carry the amino acid threonine in mitochondria. It is 62 ribonucleotides long. [2]

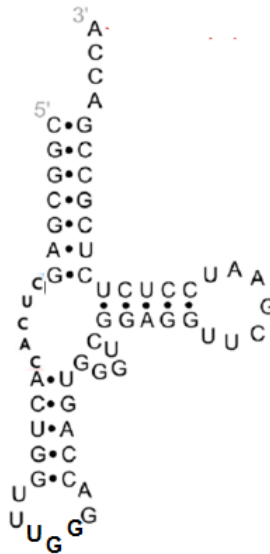


Fig. 3.2

(b) The structures of the tRNA molecules in the mitochondria and the cytoplasm of a eukaryotic cell are different although they are both able to carry the amino acid threonine. Suggest a reason for this difference.

1. Mitochondria have 70S ribosomes, while 80S ribosomes are found in the cytoplasm;
2. Different tRNA molecules have specific 3D conformation that are complementary to the different A and P sites in mitochondrial and cytoplasmic ribosomes; OR
3. They are coded for by the different genes found in the mitochondrial DNA and nuclear DNA respectively;
4. Different sequences result in the formation of hydrogen bonds at different locations;
5. AVP e.g. evolutionary reasons;

[2]

There are 61 unique tRNA molecules in the cytoplasm of a eukaryotic cell.

(c) (i) Explain why there are 61 unique tRNA molecules in the eukaryotic cell.

1. The codon / anticodon is made up of 3 nucleotides and there are 4 different bases, hence 64 possible unique permutations of triplet codes;
2. Stop codons do not code for amino acids/ bind to a release factor;
3. *There are 3 stop codons, hence this gives 61 unique codons. 61 tRNA molecules will have anticodons that can complementary base pair to these 61 unique codon sequences;

Pt 3 is compulsory to score full marks

[2]

- (ii) Scientists found that there are more than 120 gene loci coding for tRNA molecules in the nuclear DNA, many of which are duplicate copies of the same tRNA gene.

Suggest an advantage of having more than 120 gene loci coding for the 61 tRNA molecules in the eukaryotic cell.

1. Rapid synthesis of tRNA molecules;

2. If one copy of the tRNA gene is mutated, there are other / duplicate copies to ensure functional tRNA is synthesised; [1]

- (d) Explain how the structure of RNA polymerase allows for the synthesis of tRNA molecules.

1. The active site has a specific/complementary 3D conformation, and it binds to the promoter/template strand of the tRNA genes;

2. The specific 3D conformation of the active site is also specific/complementary to ribonucleotides and the DNA template;

3. Catalytic residues in the active site help to catalyse the formation of the phosphodiester bonds between the 3' OH group of the one nucleotide and 5' phosphate group of the other nucleotide;

4. Active site contains catalytic residues that break hydrogen bonds between the strands in DNA molecules to expose the template strand;

5. Ref to action of enzymes, e.g. lower activation energy, binding sites for general transcription factors; [3]

[Total: 12]

4 Fig. 4.1 shows the reproductive cycles of a temperate bacteriophage.

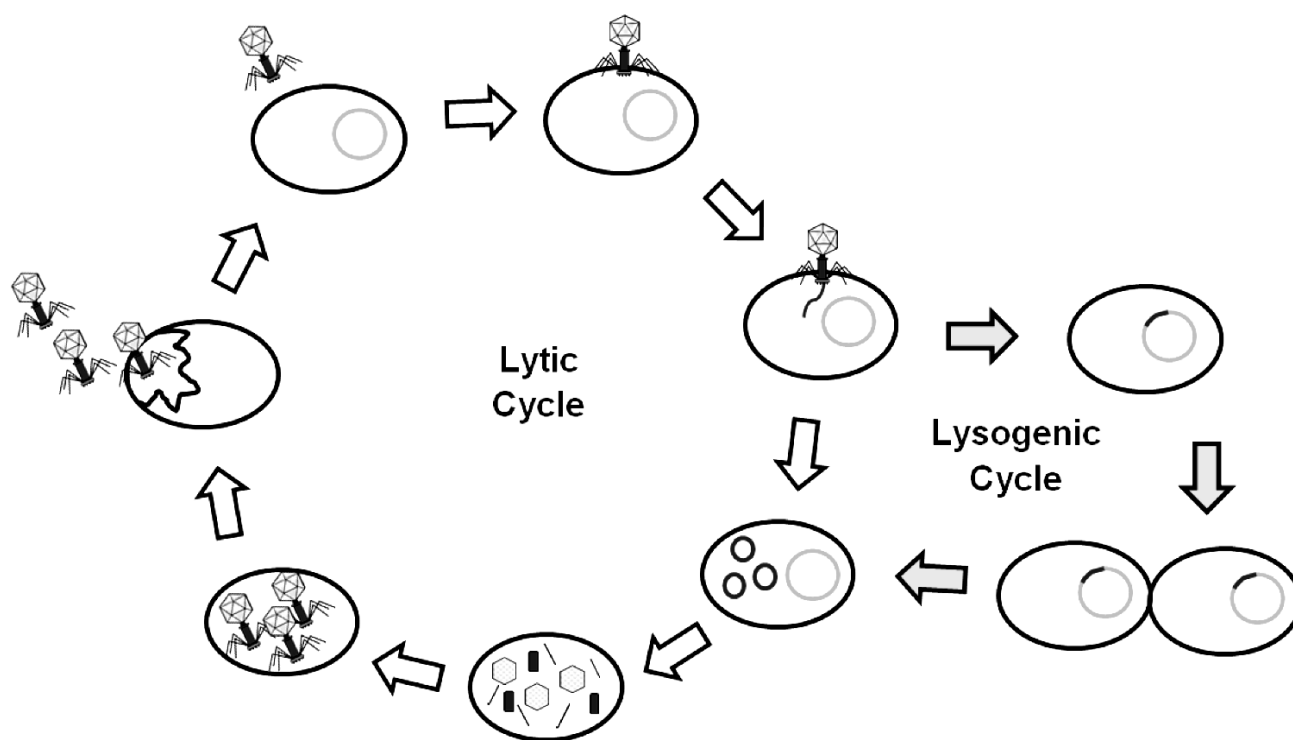


Fig. 4.1

(a) With reference to Fig. 4.1, describe the roles of the bacteriophage structures in its reproductive cycles.

1. **Tail fibres/tips - Host recognition & Adsorption:** contains proteins which bind to specific receptors on specific host bacteria for phage to attach to host;
2. **Tail sheath - Penetration & Injection:** contracts to inject phage genome/DNA into host cell;
3. **DNA/Genetic material - Lytic cycle > Phage DNA replication:** replicates to form many copies of phage genome for subsequent assembly & release of phages;
4. **DNA/Genetic material - Lytic cycle > template for synthesis of phage proteins (capsid, tail sheath and tail fibres) for subsequent assembly & release of phages;**
5. **DNA/Gen matl - Lysogenic cycle > Prophage carries viral genes which will be expressed when environment is right/replicates along with the host/integrated in host genome;**
6. **Capsid - proteins assemble around nucleic acid / carry /protect nucleic acid;**

[4]

- (b) While their prokaryotic hosts contain both DNA and RNA, bacteriophages contain only DNA. Suggest why it is sufficient for bacteriophages to contain only DNA for their reproductive cycles.

1. **Bacteriophages contain DNA as it is their genetic material/contains genes coding for phage proteins;**

2. **Phages are acellular/obligate parasites;**

3. **Host cells have RNA polymerase which transcribes phage DNA into phage mRNA;**

4. **Host cells have ribosomes/tRNAs for translation of phage mRNA into phage proteins required for maturation/assembly of phages;** [2]

- (c) Describe an advantage of the lysogenic cycle of a bacteriophage to a population of bacteria.

1. **Specialised transduction/ Prophage is inaccurately excised and so some host bacteria genes/DNA is transferred to another bacteria/ Horizontal gene transfer;**

2. **Genetic variation in the population,**

3. **Therefore higher chance of survival under different selection pressures;**

4. **AVP;**

[2]

[Total: 8]

- 5 Telomerase reverse transcriptase (TERT) is a component of the enzyme telomerase, which is responsible for maintaining telomere lengths in epidermal stem cells. These cells will differentiate to form specialised skin cells such as keratinocytes. A study on the role of the regulatory protein c-MYC in the synthesis of TERT mRNA was conducted in these cells. Fig. 5.1 shows the relative levels of TERT mRNA and c-MYC mRNA.

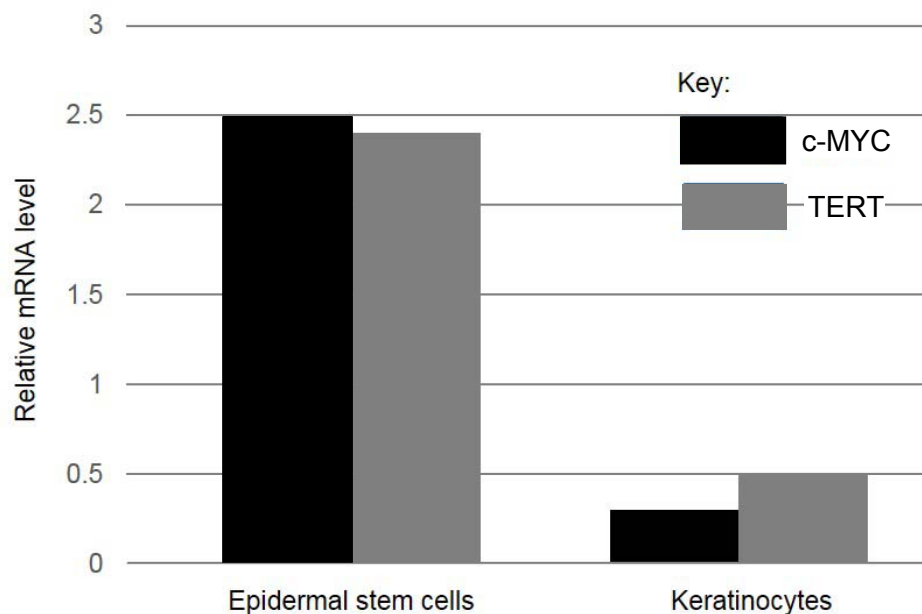


Fig. 5.1

- (a) (i) With reference to Fig. 5.1, describe the relationship between the relative levels of TERT mRNA and c-MYC mRNA.

1. **Positive/ direct relationship;**

2. **In epidermal stem cells, mRNA levels of c-Myc and TERT are both high at 2.5 and 2.4 respectively/ in keratinocytes, mRNA levels of c-Myc and TERT are both low at 0.3 and 0.5 respectively;**

[2]

- (ii) Suggest a role of the c-MYC protein in the regulation of *TERT* gene expression.

1. **c-Myc is an activator which binds to enhancer of *TERT* gene;**

2. **Ref. to recruitment of DNA bending proteins, mediator proteins and other transcription factors to form transcription initiation complex (TIC);**

3. **Allows formation of a more stable TIC/ Higher affinity of RNA polymerase to promoter and hence, high rate of transcription of TERT mRNA; OR**

4. **c-Myc as histone acetylase which adds acetyl group to histone tails;**

5. **Reduces affinity of histones for DNA, chromatin becomes less compact;**

6. **TERT gene is transcribed at a high rate; OR**

7. **c-Myc as histone demethylase which removes methyl groups from histone tails;**

8. **Proteins no longer bound to DNA, chromatin becomes less compact;**

9. **TERT gene is transcribed at a high rate;**

[3]

(b) Explain why TERT has a significant role in epidermal stem cells.

1. Ref to end replication problem;

2. Therefore telomere length will be shortened to the point where the cell undergoes programmed cell death/replicative senescence is triggered;

3. Epidermal stem cells can *continue* to divide by mitosis/self-renewal for a long period of time to form a pool of cells;

4. *TERT maintains telomere length by synthesizing multiple DNA repeats at the 3' OH end of the DNA molecule;

[3]

(c) Suggest why *TERT* gene is absent in bacteria.

1. Bacteria has circular DNA;

2. Ref to lack of end replication problem;

[2]

[Total: 10]

6 Many diseases arise due to gene mutations and/or chromosomal aberrations.

(a) Distinguish between gene mutations and chromosomal aberrations.

	Gene mutation	Chromosomal aberration
1. Definition	1a. Change in nucleotide sequence	1b. Change in chromosomal structure or number;
2. No. of genes affected	2a. Affects a single gene	2b. Usually affects multiple genes (because involves a segment or an entire chromosome hence);
3. Mechanisms	3a. Substitution, deletion or addition of nucleotides (any 2)	3b. Translocation, deletion or inversion of chromosomal structures or non-disjunction events during nuclear division (any 2);

Any 2@1m each

[2]

Fig. 6.1 shows an error occurring during meiosis II.

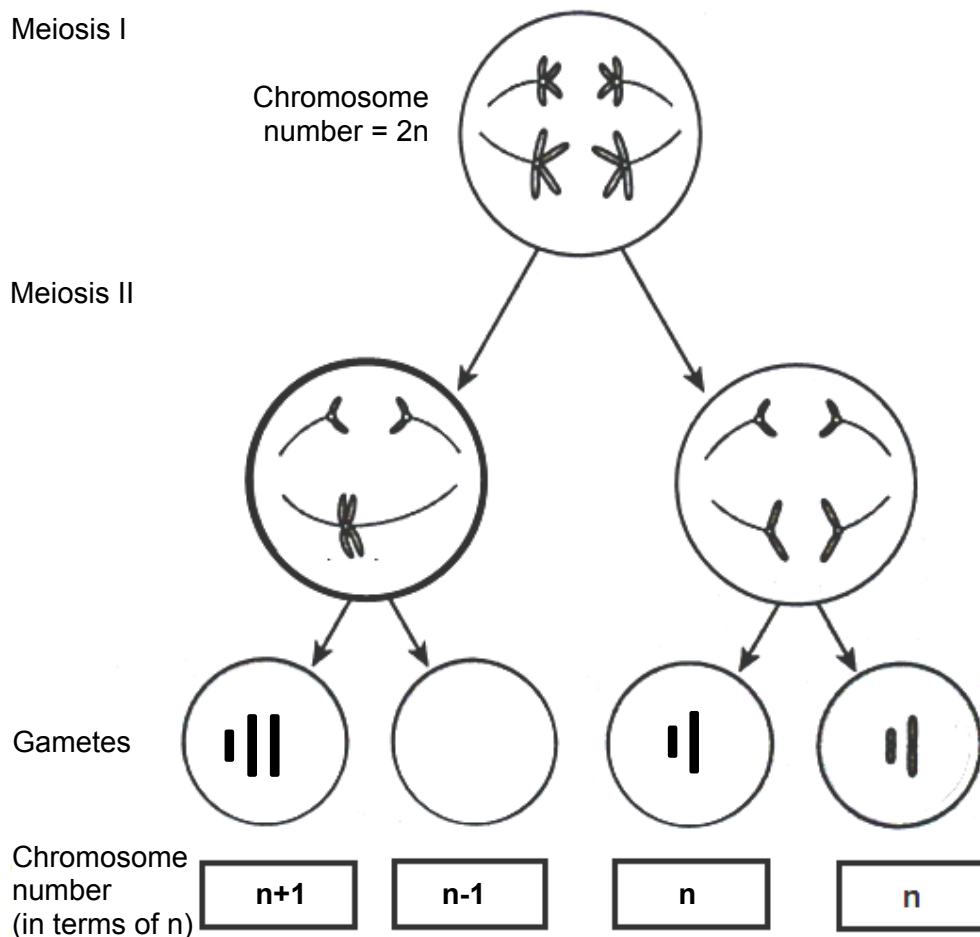


Fig. 6.1

(b) (i) Complete Fig. 6.1 with the correct chromosome structures and chromosome number (in terms of n) in the gametes. [2]

(ii) Explain the error shown in Fig. 6.1.

1. **Non-disjunction caused by the failure of sister chromatids to separate during anaphase II;**
2. **due to problems in the shortening of the spindle fibres/ centromeres fail to divide;**

[2]

(iii) Predict the chromosomal number (in terms of n) in each of the four gametes if a similar error had occurred in meiosis I instead of meiosis II.

$n+1$, $n+1$, $n-1$, $n-1$

[1]

(c) Such errors can occur during mitosis too. Comment on whether this would lead to changes in the gene pool of a population.

1. **If this occurs in somatic cells only / does not affect the germ line cells, it only affects the individual / not passed down to offspring, so no effect on overall gene pool in population; OR**
2. **If this occurs in germ line cells, it will be passed to gametes, and possibly to offspring, so gene pool of a population will be affected;**
3. **If this occurs in zygotic / embryonic stem cells, the whole individual will be affected, so gene pool of a population will be affected;**

[1]

[Total: 8]

7 (a) Explain how genotype is linked to phenotype.

1. **Genotype, the genetic makeup of an organism, is made up of genes which are sequences of deoxyribonucleotides;**
2. **Genes are transcribed to form mRNAs, and mRNAs are translated into proteins;**
3. **These proteins have particular functions/roles and may affect metabolic pathways, giving rise to the phenotype of the organism;**

[3]

In the fruit fly, *Drosophila*, a gene determines the appearance of a dark transverse band on the thorax. Another gene determines the presence of crossveins on the wings. Two pure-breeding *Drosophila* were crossed, where one is banded with crossveins on the wings while the other is non-banded without crossveins on the wings. The F1 offspring were then test-crossed to produce offspring in the following ratio:

banded without crossveins	21
banded with crossveins	483
non-banded without crossveins	512
non-banded with crossveins	35

(b) Using the symbols **B** for non-banded thorax and **b** for banded thorax, **D** for presence of crossveins and **d** for absence of crossveins, draw a genetic diagram to show the results of the test-cross.

Parental phenotypes	non-banded with crossveins		X	banded without crossveins	
Parental genotypes	(Bd)(bD)		X	(bd)(bd) ;	
Gametes	(Bd) (bD) (bd) (BD)			(bd) ;	
Offspring genotypes	(Bd)(bd)	(bD)(bd)	(BD)(bd)	(bd)(bd)	
Offspring phenotypes	non-banded without crossveins	banded with crossveins	non-banded with crossveins	banded without crossveins ;	
	512	438	35	21	
	parental		recombinant		;

[4]

(c) Explain how the new combination of alleles arose in the gametes of the test-cross in part (b).

1. Crossing over between non-sister chromatids of homologous chromosomes;

2. at the chiasma in prophase I;

3. Allele coding for non-banded thorax/absence of crossveins from one chromosome swap with corresponding allele coding for banded thorax/presence of crossveins from the other chromosome;

(Resulting in recombinant chromosomes containing genetic material from both parents)

[3]

(d) Distinguish sex linkage from the mode of inheritance of the two genes observed in *Drosophila*.

Basis of comparison	Sex linkage	Autosomal linkage
1. Location of genes	Genes are located on the sex chromosomes (mostly located on the X chromosome)	Genes are located on (the same) autosomes for autosomal linkage;
2. Distribution of phenotypes	Genes transmitted on sex chromosomes show unequal distribution of the particular trait between the sexes	Breeding experiment produce approximately equal numbers of the parental phenotypes and a significantly smaller number of phenotypes showing new combinations of characteristics, also in equal number;
3. Results of reciprocal crosses	The reciprocal crosses would result in different phenotypic ratios in the offspring	Results of reciprocal crosses would not differ

[2]

[Total: 12]

- 8 Concentrations of ribulose biphosphate (RuBP) and reduced NADP were measured in samples of actively photosynthesising green algae in an experimental chamber. CO_2 is supplied continuously into the chamber throughout the experiment.

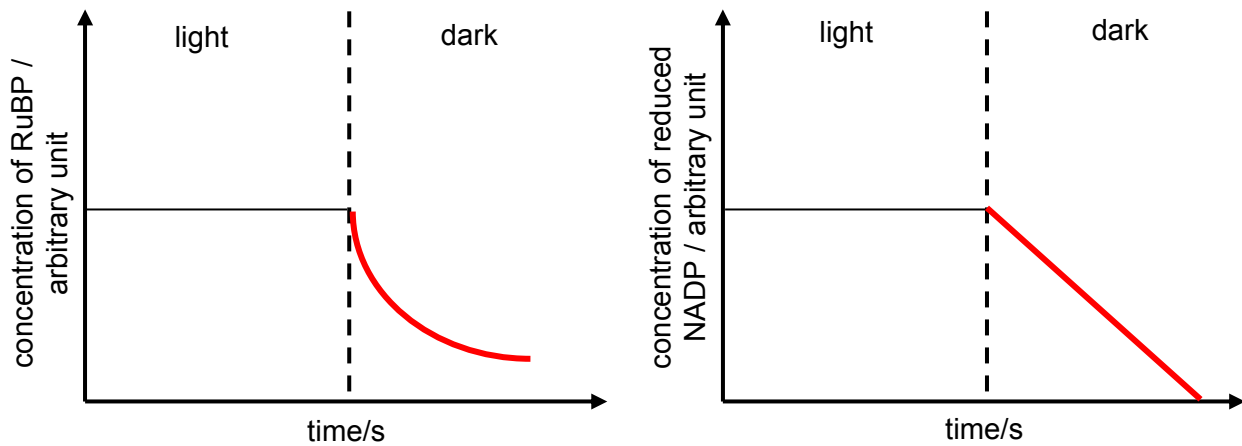


Fig. 8.1

- (a) (i) In Fig. 8.1, complete each graph to show the concentrations of RuBP and reduced NADP when the light source was turned off. [2]

- (ii) Explain the shape of the two graphs when the light source was turned off.

1. **Photophosphorylation/light dependent reaction ceased, reduced NADP and ATP is not produced;**
2. **Existing reduced NADP is used up to reduce glycerate-3-phosphate (PGA) to glyceraldehyde-3-phosphate (GALP) in the Calvin cycle/ light independent reaction;**
3. **Ribulose biphosphate (RuBP) is not regenerated due to the absence of ATP;**
4. **RuBP combines with CO_2 to form PGA/carbon fixation, and its concentration decreases;**
5. **AVP;**

[4]

(b) 'Calvin cycle is the reverse of Krebs cycle'. Comment on this statement.

1. Calvin cycle results in the reduction of substrates to form carbohydrates while Krebs cycle results in the oxidation of respiratory substrates / carbohydrates OR Calvin cycle is an anabolic process while Krebs cycle is a catabolic process;
2. Calvin cycle used CO_2 while Krebs cycle produces CO_2 ;
3. Calvin cycle uses ATP while Krebs cycle produces ATP (overall);
4. Calvin cycle uses reduced hydrogen carriers while Krebs cycle produces reduced hydrogen carriers;
5. Contrast one e.g. of enzyme in cycle (e.g. carboxylase is involved in Calvin cycle vs decarboxylase in Krebs cycle / both processes involve dehydrogenases;
6. AVP;

[3]

(c) The Krebs cycle must be carefully regulated by the cell. If it is permitted to proceed unchecked, large amounts of metabolic energy would be wasted in the over-production of reduced coenzymes and ATP. α -ketoglutarate dehydrogenase is an enzyme in the Krebs cycle which catalyses the conversion of α -ketoglutarate to succinyl coA.

Suggest how reduced NAD can inhibit the action of α -ketoglutarate dehydrogenase.

1. Excess/ accumulation of reduced NAD inhibit the action of α -ketoglutarate dehydrogenase by end product inhibition;
 2. Reduced NAD binds to the active site on α -ketoglutarate dehydrogenase (as it is a co-enzyme);
 3. It competes with NAD^+ for the active site, so oxidative decarboxylation of α -ketoglutarate cannot occur to form succinate, Krebs cycle stops / slows down;
- OR
2. Reduced NAD binds to the allosteric site of the enzyme;
 3. Changing the 3D configuration of the active site of enzyme, so oxidative decarboxylation of α -ketoglutarate cannot occur to form succinate;

[3]

[Total: 12]

- 9 A study reported on the change in beetle sizes in a population due to climate change. Larger beetles are affected when compared to smaller beetles. This could be because larger beetles are less likely to get enough oxygen to sustain a higher metabolic rate caused by higher atmospheric temperatures. Fig. 9.1 shows the effect of an increase of 2°C in temperature on the size of larger and smaller beetles in a population over 100 generations.

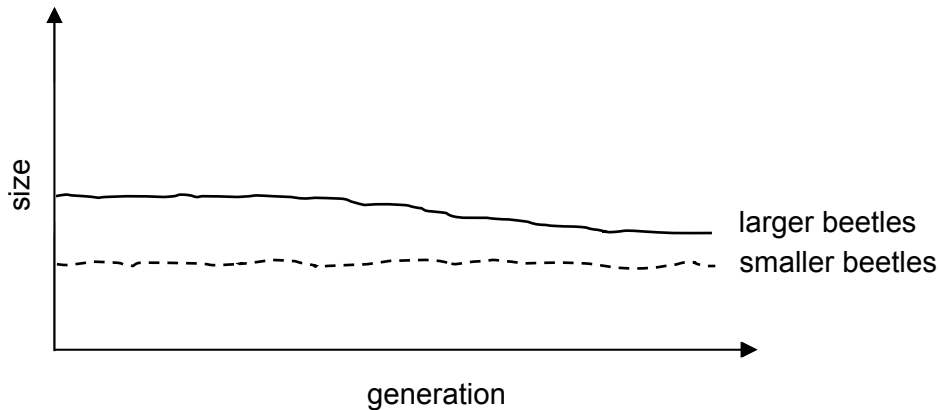


Fig. 9.1

- (a) Explain how a temperature rise of 2°C may cause a change in the gene pool of the population of beetles over time.

1. **Genetic variation in population expressed as phenotypic variation, i.e. range of sizes;**
2. **Higher temperature is the selection pressure;**
3. **Smaller beetles are selected for because they are more likely to... (ref to preamble);**
4. **Resulting in more of these surviving and reproducing and passing on alleles coding for small size to their offspring;**
5. **Over time, gene pool changes where there is a higher frequency of alleles coding for smaller sizes in the population;**

[4]

- (b) Suggest why a speciation event is unlikely to occur in this population of beetles as a result of climate change.

1. **No reproductive isolation between the 2 beetle phenotypes / there is gene flow within the population;**
2. **No accumulation of different mutations, i.e no new diff in behaviour, no new diff in size of copulatory organs, no new difference in pheromones;**
3. **Less differences in beetle size in the population, resulting in a more homogenous population and less genetic differences;**

[2]

(c) Suggest the consequences of a decrease in beetle size on the ecosystem.

1. **Beetles are preyed upon, so less food for animals higher up in food chain;**
2. **Beetles act as detritivores, smaller beetles mean slower rate of decomposition and hence recycling of nutrients;**
3. **Beetles can be pollinators of plants, so lower rate of pollination leads to reduction in certain plant populations/loss of certain plant species;**
4. **Ref to biodiversity;**

[2]

(d) Describe how molecular methods can be used to confirm that two beetles with similar morphology are of the same species.

1. **Extract/Obtain DNA from both beetles;**
2. **Amplify it via polymerase chain reaction;**
3. **Use the same restriction enzyme to cut the gene/DNA sequence to result in different fragment lengths (optional) / RFLP analysis**
OR
Designing specific primers that bind to the ends of a target gene/sequence;
4. **Subject it to gel electrophoresis to separate DNA fragments by size**
OR
separate DNA segments by size;
5. **Conduct autoradiography with single-stranded, radioactive probe to confirm presence of target sequences in the fragments/segments;**
6. ***Same/similar banding pattern for both beetles → same species;**
7. **AVP, e.g. DNA-DNA hybridization, DNA sequencing;**

Pt 6 is a compulsory point to get full marks

[4]

[Total: 12]

- 10** Multidrug-resistant tuberculosis (MDR-TB) is a form of tuberculosis (TB) caused by the bacterium *Mycobacterium tuberculosis*, which is resistant to treatment with the antibiotic rifampin. The first antibiotic treatment for TB began in 1943 and some strains *M. tuberculosis* developed resistance to rifampin through genetic changes. Fig. 10.1 shows rifampin binding to RNA polymerase in bacterial cells.

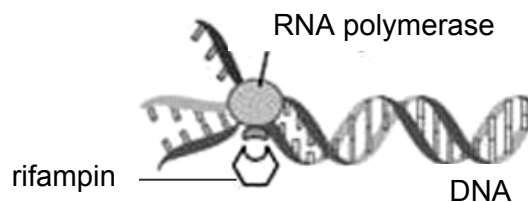


Fig. 10.1

- (a) With reference to Fig. 10.1, explain why rifampin may be used to treat tuberculosis.
1. **RNA polymerase inhibited in bacterial cells and hence, no transcription/ expression of genes;**
 2. **Proteins/ enzymes crucial for cellular processes name 1 (e.g. no enzymes for respiration) absent, , hence resulting in bacterial cell death;**
 3. **Rifampin does not bind to eukaryotic RNA polymerase so host cells are not affected;**

[2]

- (b) Suggest how the genetic changes which caused rifampin resistance could be transferred within a population of *M. tuberculosis* through direct contact of bacteria cells.
1. **bacterial conjugation;**

2. **F⁺ cell/donor bacterial cell with F factor produces sex pilus to attach itself to F⁻ cell/recipient cell, forming a mating bridge/conjugation tube between the two bacterial cells;**
3. **which allows F⁺ cell to transfer its F plasmid containing the antibiotic resistance gene to the F⁻ cell (by rolling circle mechanism);**

[3]

Bacilli Calmette-Guérin (BCG) is a vaccine containing attenuated bacteria and can provide protection against tuberculosis. The attenuated bacteria have antigens which stimulate an immune response. Recent studies have shown that BCG can also give protection against leprosy, a disease caused by the bacterium *Mycobacterium leprae*.

(c) (i) Outline the roles of T lymphocytes in the adaptive immune response after the administration of BCG.

1. Naïve helper T cells with specific/ complementary T cell receptors bind to the MHC-peptide complexes and are activated by the cytokines released from antigen-presenting cells;

2. Activated helper T cells in turn (releases cytokines) which activate naïve B cells/ cytotoxic T cells;

3. B cells/ (effector cells) secrete antibodies to perform opsonisation/ agglutination/activation of complement system

OR

Cytotoxic T cells secrete proteins (perforin/ granzymes) to kill infected cells;

4. Formation of memory T/B cells at the end of primary immune response;

[3]

(ii) Suggest why BCG can also provide protection against leprosy.

1. *Bacteria which causes leprosy has antigens/epitopes with similar 3D conformation as those in *Mycobacterium tuberculosis*;

2. Receptors of memory T and B cells can bind to antigens/epitopes of *Mycobacterium leprae*;

3. AVP;

[2]

[Total: 10]

